MIL-P-10388G
20 April 1982
SUPERSEDING
MIL-P-10388F
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MILITARY SPECIFICATION

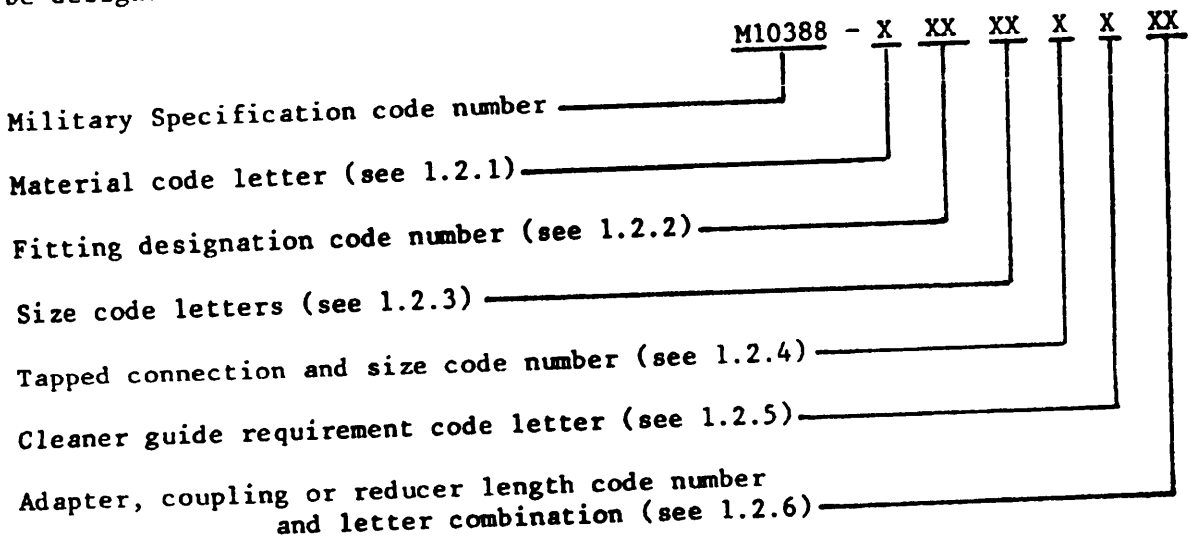
PIPE FITTINGS: ONE OR MORE ENDS GROOVED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers grooved- and combination-type pipe fittings.

1.2 Classification. Pipe fittings covered by this specification shall be designated in the following form (see 6.2 and 6.6).



Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4710
FSC 4730

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1.2.1 Material. The material of the fitting is identified by a single letter (see table I).

TABLE I. Type material code letter.

Code Letter	Material
A	Aluminum alloy
B	Malleable iron, ductile iron or steel.

1.2.2 Fitting designation. The kind of fitting is identified by a two digit number (see table II).

TABLE II. Fitting designation code number.

Fitting kind	Fitting code number	Fitting kind	Fitting code number
ADAPTERS		ELBOWS	
Flanged (class 125) one end	01	90 degree both ends grooved	11
Flanged (class 150) one end	02	90 degree external threaded one end	12
Flanged (class 250) one end	03	45 degree both ends grooved	13
Flanged (class 300) one end	04	45 degree external threaded one end	14
CAP	05	22-1/2 degree both ends grooved	15
COUPLINGS		11-1/4 degree both ends grooved	16
Beveled one end	06	Adjustable both ends grooved	17
External threaded one end	07		
Grooved both ends	08	LATERALS	
Internal threaded one end	09	45 degree all ends grooved	18
CROSS	10		

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TABLE II. Fitting designation code number. - Continued

Fitting kind	Fitting code number	Fitting kind	Fitting code number
REDUCERS		Flanged (class 300) small end	27
Grooved both ends	19	Threaded small end	28
Flanged (class 125) large end	20		
Flanged (class 150) large end	21	TEES	
Flanged (class 250) large end	22	Grooved all ends	29
Flanged (class 300) large end	23	Reducing (grooved all ends)	30
Flanged (class 125) small end	24	Reducing (run ends grooved, branch end external threaded)	31
Flanged (class 150) small end	25	Reducing (run ends grooved, branch end internal threaded)	32
Flanged (class 250) small end	26	UNION (flexible) both ends grooved	33

1.2.3 Size. The nominal pipe size(s) of the fitting is identified by two letters. Reducer sizes are covered in table IV. Tee sizes are covered in table V. Sizes for all other fittings are covered in table III.

TABLE III. Size code letters (except reducers and tees).

Nominal pipe size (inches)	Size code letters	Nominal pipe size (inches)	Size code letters
3/4	AA	4	AI
1	AB	5	AJ
1-1/4	AC	6	AK
1-1/2	AD	8	AL
2	AE	10	AM
2-1/2	AF	12	AN
3	AG	14	AO
3-1/2	AH		

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TABLE IV. Reducer size code letters.

Nominal pipe size (inches)	Size code letters	Nominal pipe size (inches)	Size code letters
1-1/2 x 1	BA	5 x 4	BP
2 x 3/4	BB	6 x 1	BQ
x 1	BC	x 2	BR
x 1-1/4	BD	x 3	BS
x 1-1/2	BE	x 4	BT
2-1/2 x 2	BF	x 5	BU
3 x 1	BG	8 x 3	BV
x 2	BH	x 4	BW
x 2-1/2	BI	x 5	BX
3-1/2 x 3	BJ	x 6	BY
4 x 1	BK	10 x 4	BZ
x 2	BL	x 6	CA
x 2-1/2	BM	x 8	CB
x 3	BN	12 x 6	CD
x 3-1/2	BO	x 8	CE
		x 10	CF

TABLE V. Tee size code letters.

Nominal pipe size (inches)	Size code letters	Nominal pipe size (inches)	Size code letters	Nominal pipe size (inches)	Size code letters
3/4 x 3/4 x 3/4	DA	4 x 4 x 3	DS	10 x 10 x 2	EK
1 x 1 x 1	DB	x 4	DT	x 3	EL
1-1/4 x 1-1/4 x 1-1/4	DC	5 x 5 x 2	DU	x 4	EM
1-1/2 x 1-1/2 x 1-1/2	DD	x 3	DV	x 5	EN
2 x 2 x 3/4	DE	x 4	DW	x 6	EO
x 1	DF	x 5	DX	x 8	EP
x 1-1/2	DG	6 x 6 x 2	DY	x 10	EQ
x 2	DH	x 2-1/2	DZ	12 x 12 x 3	ER
2-1/2 x 2-1/2 x 2-1/2	DI	x 3	EA	x 4	ES
3 x 3 x 1	DJ	x 4	EB	x 5	ET
x 1-1/2	DK	x 5	EC	x 6	EU
x 2	DL	x 6	ED	x 8	EV
x 2-1/2	DM	8 x 8 x 2	EE	x 10	EW
x 3	DN	x 3	EF	x 12	EX
3-1/2 x 3-1/2 x 3-1/2	DO	x 4	EG	14 x 14 x 8	EY
4 x 4 x 1	DP	x 5	EH	x 10	EZ
x 2	DQ	x 6	EI	x 12	FA
x 2-1/2	DR	x 8	EJ	x 14	FB

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1.2.4 Tapped connection and size. Whether a fitting has a tapped connection and the size of the drilled and tap connection are identified by a single number (see table VI).

TABLE VI. Tapped connection and size code number.

Code number	Size of drilled and tapped connection
1	None
2	1/4 inch
3	1/2 inch
4	3/4 inch
5	2 inch

1.2.5 Cleaner Guide. Whether tees are furnished with a cleaner guide is identified by a single letter (see table VII).

TABLE VII. Cleaner guide code letter.

Code letter	Cleaner guide requirement
A	Without cleaner guide (for tees)
B	With cleaner guide (for tees)
C	Without cleaner guide (for fittings other than tee)

1.2.6 Adapter, coupling and reducer length. The length of the adapter, coupling or reducer shall be identified by a number and letter combination. The number shall be the number of whole inches in the required length and the letter code shall indicate the fractional part of an inch of the required length (example: 8 inches long is 8A, 5-1/2 inches long is 5C, 3/4 inch long is OD and 10-1/4 inches long is 10B) (see table VIII).

TABLE VIII. Fractional inch code letter.

Code Letter	Fractional Inch
A	0
B	1/4
C	1/2
D	3/4

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-T-60	- Tape: Packaging, Waterproof.

MILITARY

MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-T-704	- Treatment and Painting of Material.
MIL-P-11087	- Pipe, Steel: Grooved, Threaded, or Plain Ends.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASTM A 47	- Malleable Iron Castings.
ANSI/ASTM A 536	- Ductile Iron Castings.
ANSI/ASTM B 26	- Aluminum-Alloy Sand Castings.
ANSI/ASTM B 210	- Aluminum-Alloy Drawn Seamless Tubes.
ANSI/ASTM B 247	- Aluminum-Alloy Die and Hand Forgings.
ANSI B2.1	- Pipe Threads (Except Dryseal).
ANSI B16.1	- Cast-Iron Pipe Flanges and Flanged Fittings (Class 25, 125, 250, and 800).
ANSI B16.25	- Butt Welding Ends.
ANSI B16.5	- Steel Pipe Flanges, and Flanged Fittings (Including ratings for Class 150, 300, 400, 600, 900, 1500 and 2500).
ANSI B46.1	- Surface Texture (Surface Roughness, Waviness and Lay).

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. Fittings shall be malleable iron, ductile iron, steel, or aluminum alloy, and shall be furnished with various combinations of flanged, threaded, beveled, and grooved ends. All fittings shall have one or more ends grooved.

3.2 First article (preproduction model). The contractor shall furnish one or more pipe fittings for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods and choice of design detail will produce fittings that comply with the requirements of this specification. Examination and tests shall be as specified in section 4 and shall be subject to surveillance and approval be the Government (see 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all the provisions of this specification (see 6.5).

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3.3.1 Adapters and flanged reducers.

3.3.1.1 Iron flanged fittings. Iron flanged fittings shall be fabricated from steel pipe conforming to MIL-P-11087, size and weight as shown in table IX, and flanges of malleable iron castings conforming to ANSI/ASTM A 47, grade 32510, or ductile iron castings conforming to ANSI/ASTM A 536, grade 65-45-12.

3.3.1.2 Steel flanged fittings. Steel flanged fittings shall be fabricated from steel pipe conforming to MIL-P-11087, size and weight as shown in table IX, and flanges of steel selected by the contractor subject to all the provisions of this specification.

3.3.1.3 Aluminum flanged fittings. Aluminum flanged fittings shall be fabricated from round aluminum-alloy tubing conforming to ANSI/ASTM B 210, alloy UNS A96061, T6, wall thickness to fit the application, and flanges of aluminum-alloy forgings conforming to ANSI/ASTM B 247 alloy UNS A96151, or aluminum-alloy castings conforming to ANSI/ASTM B 26, alloy and temper to fit the application.

3.3.2 Couplings.

3.3.2.1 Steel couplings. Steel couplings shall be steel pipe conforming to MIL-P-11087, size and weight as shown in table IX.

3.3.2.2 Aluminum couplings. Aluminum couplings shall be round aluminum-alloy tube conforming to ANSI/ASTM B 210, alloy UNS A96061, T6, wall thickness to fit the application.

TABLE IX. Diameter and weight.

Nominal pipe size (inches)	Outside pipe diameter (inches)	Minimum weight (lb/ft)
3/4	1.050	1.13
1	1.315	1.68
1-1/4	1.660	2.27
1-1/2	1.900	2.72
2	2.375	3.65
2-1/2	2.875	5.79
3	3.500	7.58
3-1/2	4.000	9.11
4	4.500	10.79
5	5.563	14.62
6	6.625	18.97
8	8.625	22.36
10	10.750	28.04
12	12.750	33.38
14	14.000	36.71

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3.3.3 Fittings (except adapter, coupling, and flanged reducer).

3.3.3.1 Iron fittings. Fittings shall be malleable iron castings conforming to ANSI/ASTM A 47, grade 32510, or ductile iron castings conforming to ANSI/ASTM A 536, grade 65-45-12.

3.3.3.2 Steel fittings. Steel fittings shall be steel selected by the contractor and subject to all provisions of this specification.

3.3.3.3 Aluminum fittings. Aluminum fittings shall be aluminum-alloy forgings conforming to ANSI/ASTM B 247, alloy UNS A96151, or aluminum-alloy castings conforming to ANSI/ASTM B 26, alloy and temper to fit the application.

3.4 Design. Fittings shall be malleable iron, ductile iron, steel, or aluminum alloy, as specified (see 1.2.1). All fittings shall have one end grooved and the other end(s) beveled, flanged, grooved, or threaded. The groove may be continuous or backstop lug design. When specified (see 1.2.4), fittings shall be drilled, tapped, and a pipe plug inserted for 1/4 inch, 1/2 inch, 3/4 inch, or 2 inch NPT connection.

3.4.1 Adapters and flanged reducers. Cast iron adapters and flanged reducers shall consist of steel pipe threaded into a cast iron flange. Steel adapters and flanged reducers shall consist of steel pipe threaded or welded to a steel flange. When pipe and flange are threaded together, the threads shall be coated with a nonhardening, nongalling compound inert to aromatic gasoline. Aluminum adapters and flanged reducers shall consist of aluminum-alloy tubing welded to an aluminum-alloy flange.

3.4.2 Tees. When specified (see 1.2.5), tees shall be furnished with a 1/2-inch steel cleaner guide, brazed in a horizontal position across the inside diameter of the branch outlet, with the inner side in line with the bore of the main outlets. The brazing operation shall not reduce the strength of the fitting.

3.5 Dimensions.

3.5.1 Fitting lengths. Fitting dimensions shall conform to table X for the size specified. Lengths are not given in table X for adapters, couplings and reducers. Therefore, adapter, coupling and reducer lengths must be specified as part of the military specification part number (see 1.2.6).

3.5.2 Fitting ends.

3.5.2.1 Beveled. Beveled ends shall be in accordance with ANSI B16.25.

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TABLE X. Fitting length.

<table>
<tr><th colspan="5">Adapters - one end flanged.</th></tr>
<tr><th rowspan="2">Nominal size (inches)</th><th colspan="4">Length of adapter (inches) measured end to end</th></tr>
<tr><th>Class 125</th><th>Class 150</th><th>Class 250</th><th>Class 300</th></tr>
<tr><td>3/4
1
1-1/4
1-1/2
2
2-1/2
3
3-1/2
4
5
6
8
10
12
14</td><td colspan="4">Length for all sizes and designations of flanged adapters shall be as specified (see 1.2.6).</td></tr>
<tr><th colspan="5">Cap</th></tr>
<tr><td colspan="5">Nominal size (inches) 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 3-1/2, 4, 5, 6, 8, 10, 12, and 14.</td></tr>
<tr><td colspan="5">Length: Seats flush with outside of coupling.</td></tr>
<tr><th colspan="5">Couplings</th></tr>
<tr><th rowspan="3">Nominal size (inches)</th><th colspan="4">Length of coupling (inches) measured end to end</th></tr>
<tr><th rowspan="2">Beveled one end</th><th colspan="2">Threaded one end</th><th rowspan="2">Grooved both ends</th></tr>
<tr><th>External</th><th>Internal</th></tr>
<tr><td>3/4
1
1-1/4
1-1/2
2
2-1/2
3
3-1/2
4
5
6
8
10
12
14</td><td colspan="4">Length for all sizes and designations of coupling shall be as specified (see 1.2.6).</td></tr>
</table>

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TABLE X. Fitting length. - Continued

Cross	
Nominal size (inches)	Length of cross (inches) measured center to ends, all ends grooved
3/4	2-1/4
1	2-1/4
1-1/4	2-3/4
1-1/2	2-3/4
2	3-1/4
2-1/2	3-3/4
3	4-1/4
3-1/2	4-1/2
4	5
5	5-1/2
6	6-1/2
8	7-3/4
10	9
12	10
14	11

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TABLE X. Fitting length. - Continued

Elbows									
Nominal size (inches)	Length of elbows (inches) Both ends grooved					External threaded one end		Center to grooved end	
	Center to end				End to end	Center to threaded end			
	90°	45°	22-1/2°	11-1/4°	Adjustable	90°	45°	90°	45°
3/4	2-1/4	1-1/2	1-5/8	1-3/8	-	2-1/4	1-1/2	2-1/4	1-1/2
1	2-1/4	1-3/4	1-5/8	1-3/8	3-1/4	2-1/4	1-3/4	2-1/4	1-3/4
1-1/4	2-3/4	1-3/4	1-3/4	1-3/8	-	2-3/4	1-3/4	2-3/4	1-3/4
1-1/2	2-3/4	1-3/4	1-3/4	1-3/8	-	2-3/4	1-3/4	2-3/4	1-3/4
2	3-1/4	2	1-7/8	1-3/8	3-3/4	4-1/4	3	3-1/4	2
2-1/2	3-3/4	2-1/4	2	1-1/2	4	3-3/4	2-1/4	3-3/4	2-1/4
3	4-1/4	2-1/2	2-1/4	1-1/2	4-1/2	6	4-1/4	4-1/4	2-1/2
3-1/2	4-1/2	2-3/4	2-1/2	1-3/4	5	6-1/4	2-3/4	4-1/2	2-3/4
4	5	3	2-5/8	1-3/4	5-1/4	7-1/4	5-1/4	5	3
5	5-1/2	3-1/4	2-7/8	2	-	-	-	-	-
6	6-1/2	3-1/2	3-1/8	2	6-1/4	6-1/2	3-1/2	6-1/2	3-1/2
8	7-3/4	4-1/4	3-7/8	2	7-3/4	-	-	-	-
10	9	4-3/4	4-3/8	2-1/8	-	-	-	-	-
12	10	5-1/4	4-7/8	2-1/4	-	-	-	-	-
14	21	8-3/4	5	3-1/2	-	-	-	-	-

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TABLE X. Fitting length. - Continued.

45° Lateral		
Nominal size (inches)	Length of lateral (inches) All ends grooved	
	Center to long end	Center to short end
3/4	4-1/2	2
1	5	2-1/4
1-1/4	5-3/4	2-1/2
1-1/2	6-1/4	2-3/4
2	7	2-3/4
2-1/2	7-3/4	3
3	8-1/2	3-1/4
3-1/2	10	3-1/2
4	10-1/2	3-3/4
5	12-1/2	4
6	14	4-1/2
8	18	6
10	20-1/2	6-1/2
12	23	7
14	26-1/2	7-1/2

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TABLE X. Fitting length. - Continued

Reducers				
Nominal size (inches)	Length of reducers (inches) measured end to end			
	Small end threaded	Small end flanged	Large end flanged	Both ends grooved
1-1/2 x 1 2 x 3/4 2 x 1 2 x 1-1/4 2 x 1-1/2 2-1/2 x 2 3 x 1 3 x 2 3 x 2-1/2 3-1/2 x 3 4 x 1 4 x 2 4 x 2-1/2 4 x 3 4 x 3-1/2 5 x 4 6 x 1 6 x 2 6 x 3 6 x 4 6 x 5 8 x 3 8 x 4 8 x 5 8 x 6 10 x 4 10 x 6 10 x 8 12 x 6 12 x 8 12 x 10	Length for all sizes and designations of reducers shall be as specified (see 1.2.6).			

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TABLE X. Fitting length. - Continued

Tees		
Nominal size (inches)	Length of tees (inches) measured center to ends	
	Grooved all ends	Reducing
3/4 x 3/4 x 3/4	2-1/4	-
1 x 1 x 1	2-1/4	-
1-1/4 x 1-1/4 x 1-1/4	2-3/4	-
1-1/2 x 1-1/2 x 1-1/2	2-3/4	-
2 x 2 x 3/4	-	3-1/4
x 1	-	3-1/4
x 1-1/2	-	3-1/4
x 2	3-1/4	-
2-1/2 x 2-1/2 x 2-1/2	3-3/4	-
3 x 3 x 1	-	4-1/4
x 1-1/2	-	4-1/4
x 2	-	4-1/4
x 2-1/2	-	4-1/4
x 3	4-1/4	-
3-1/2 x 3-1/2 x 3-1/2	4-1/2	-
4 x 4 x 1	-	3-3/4
x 2	-	5
x 2-1/2	-	5
x 3	-	5
x 4	5	-
5 x 5 x 2	-	5-1/2
x 3	-	5-1/2
x 4	-	5-1/2
x 5	5-1/2	-
6 x 6 x 2	-	6-1/2
x 2-1/2	-	6-1/2
x 3	-	6-1/2
x 4	-	6-1/2
x 5	-	6-1/2
x 6	6-1/2	-
8 x 8 x 2	-	7-3/4
x 3	-	7-3/4
x 4	-	7-3/4
x 5	-	7-3/4
x 6	-	7-3/4
x 8	7-3/4	-

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TABLE X. Fitting length. - Continued

Tees - Continued				
Nominal size (inches)			Length of tees (inches) measured center to ends	
			Grooved all ends	Reducing
10	x 10	x 2	-	9
		x 3	-	9
		x 4	-	9
		x 5	-	9
		x 6	-	9
		x 8	-	9
		x 10	9	-
12	x 12	x 3	-	10
		x 4	-	10
		x 5	-	10
		x 6	-	10
		x 8	-	10
		x 10	-	10
		x 12	10	-
14	x 14	x 8	-	11
		x 10	-	11
		x 12	-	11
		x 14	11	-

Union (flexible)	
Nominal size (inches)	Length of union (inches) measured end to end both ends grooved
3/4	4
1	4-1/8
1-1/4	4-5/8
1-1/2	4-5/8
2	5
3	5-1/2
4	6-1/2

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3.5.2.2 Flanged.

3.5.2.2.1 Cast iron. Cast iron flanged ends shall conform to ANSI B16.1, class 125 or 250.

3.5.2.2.2 Steel. Steel flanged ends shall conform to ANSI B16.5, class 150 or 300.

3.5.2.2.3 Aluminum. Aluminum-alloy flanged ends shall be faced and drilled in accordance with ANSI B16.1, class 125 or 250, or ANSI B16.5, class 150 or 300.

3.5.2.3 Grooved. Grooved ends shall conform to the dimensions shown on figure 1. The land-sealing surface shall be smooth and free from mill marks and shall have a roughness height rating of not more than 500 in accordance with ANSI B46.1.

3.5.2.4 Threaded. Threaded ends shall be NPT in accordance with ANSI B2.1.

3.6 Working and test pressures.

3.6.1 Adapter and flanged reducers. Adapter and flanged reducers shall withstand the applicable working and hydrostatic pressures shown in table XI without evidence of leakage, rupture, or yielding.

TABLE XI. Working and hydrostatic test pressures.

Class	Maximum working pressure (WOG)1/ (psi)	Hydrostatic test pressure (psi)
125	175	250
150	230	500
250	500	900
300	720	1100

1/ Water, oil, or gas

3.6.2 Fittings (except adapter and flanged reducers). All fittings except adapter and flanged reducers shall have a working pressure of not less than 600 pounds per square inch (psi) for water, oil, or gas and shall withstand the hydrostatic pressures specified in table XII without evidence of leakage, rupture, or yielding.

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TABLE XII. Hydrostatic test pressures.

Nominal pipe size (inches)	Hydrostatic test pressure (psi)
3/4	2000
1	2000
1-1/4	2000
1-1/2	2000
2	2000
2-1/2	1800
3	1800
3-1/2	1800
4	1600
5	1600
6	1400
8	1200
10	1000
12	1000
14	800

3.7 Identification marking. Fittings shall be identified in accordance with MIL-STD-130.

3.8 Treatment and painting. Each fitting shall be cleaned, treated, and painted in accordance with the contractor's best commercial practice.

3.9 Workmanship. The fittings shall be free from pits, cracks, burrs, and any other defect that could impair serviceablity.

3.9.1 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loadings. Aluminum welding shall be done by the inert gas (argon or helium) shielded arc method. Welds may be ground, filed, or buffed, but shall not be hammered.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Preproduction inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The fittings shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The fittings shall be tested as specified in 4.5.2. Failure to pass the test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling.

4.4.1.1 For examination. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.1.2 For test. Sampling for test shall be in accordance with MIL-STD-105, inspection level S-3.

4.4.2 Examination. Samples selected in accordance with 4.4.1.1 shall be examined in accordance with 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Test. Samples selected in accordance with 4.4.1.2 shall be tested in accordance with 4.5.2. AQL shall be 4.0 percent defective. Any fitting failing to pass the test shall be considered defective.

4.5 Inspection procedure.

4.5.1 Examination. The fittings shall be examined as specified herein for the following defects:

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Major

101. Materials not as specified.
102. Dimensions not as specified.
103. Couplings, adapters, and reducers not as specified.
104. Tapped holes not as specified.
105. Flanges not as specified.
106. Beveled ends not as specified.
107. Threads not as specified.
108. Grooved ends not as specified.
109. Land-sealing surfaces not as specified.

Minor

201. Cleaner guides misalined.
202. Pipe plugs missing.
203. Identification marking missing, incomplete, or illegible.
204. Treatment and painting not as specified.
205. Workmanship not as specified.

4.5.2 Test.

4.5.2.1 Hydrostatic pressure. The fitting shall be prepared for hydrostatic test by closing all openings with blank ends. One of the blank ends shall be equipped with a nipple or adapter through which recirculated fresh water containing a rust preventive can be pumped. The fitting shall be subjected to the applicable pressure as shown on tables XI and XII for not less than 10 minutes. During this time, the fitting shall be examined. Any evidence of leakage, yielding, or rupture shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following major defects. AQL shall be 2.5 percent defective.

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110. Material not as specified. Each incorrect material shall constitute one defect.
111. Land-sealing surface of grooved end fittings not completely covered with barrier material and tape as specified.
112. Application of barrier material and tape not in accordance with figure 2 as specified.
113. Fittings of unlike size or description packed together.
114. Boxes not as specified.
115. Strapping not as specified.
116. Marking illegible, incomplete, incorrect, or missing.

5. PACKAGING

5.1 Preservation. The land-sealing surface of grooved end fittings shall be completely covered with a single wrap of barrier material conforming to MIL-B-121, type II, class 2 (see figure 2). The wrap for the land-sealing surface shall be flush with the end of the fitting and shall extend back beyond the groove not less than 1/4 inch. The wrap of the barrier material shall be covered with a single wrap of tape conforming to PPP-T-60, type IV. The tape shall be of a width that will completely cover the barrier material and shall extend not less than 1/8 inch beyond the end of the fitting and extend back beyond the barrier material not less than 1/2 inch. The end of the tape extending beyond the end of the fitting, and the tape extending beyond the wrap shall be rolled in place against the end wall of the fitting and against the fitting (see figure 2).

5.2 Packing. Packing shall be level A, level B, or commercial, as specified (see 6.2).

5.2.1 Level A. Fittings of like size and description shall be packed together in close-fitting boxes conforming to PPP-B-621, class 2, style optional or PPP-B-601, overseas type, style optional. Strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be finish B. When specified (see 6.2), strapping shall be finish A.

5.2.2 Level B. Fittings of like size and description shall be packed together in close-fitting boxes conforming to PPP-B-621, class 1, style optional, PPP-B-601, domestic type, style optional, or PPP-B-640, class 2, style optional. Strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable, finish A.

5.2.3 Commercial. The fittings shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B shall be in accordance with MIL-STD-129.

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5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. Grooved-type pipe fittings are intended for use as components of portable pipelines conveying light petroleum products or water. Couplings for grooved-type pipe are described in MIL-C-10387.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Military specification part number required (see 1.2, 3.3, 3.4, 3.4.2, 3.5.1, and 6.6).

c. Time frame required for submission of preproduction model and number of preproduction models required (see 3.2).

d. Degree of packing required (see 5.2).

e. When other than finish B strapping is required (see 5.2.1).

6.3 Preproduction model. Any changes or deviations of production fittings from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish fittings conforming to this specification.

6.4 International standardization agreement. The dimensions and tolerances shown in figure 1 are the subject of international standardization agreement (NATO STANAG 2761 and QSTAG-240). When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

6.5 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification.

6.6 Definitive military specification part number. The military specification part number corresponds to the designation and size of fitting covered by this specification and defines the requirements of the options presented under this specification. The military specification part number is a definitive part number which is formed by combining the military specification code number (M10388 for MIL-P-10388) with a dash after it; the material code letter; the fitting designation code number; the size code letter; the tapped connection and size code number; the cleaner guide requirement code letter; and the adapter, coupling or reducer length code number and letter combination as shown in paragraph 1.2.

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6.7 Materiel deterioration prevention and control (MADPAC). The contracting officer should consider for inclusion in all applicable invitations for bid, requests for proposal, and contracts the following statement:

"The technical proposal shall provide assurance that all system/item parts shall be fabricated from compatible materials, suitably finished, to provide protection from corrosion and other forms of materiel deterioration. The assurance shall be in the form of a finish plan, utilizing state-of-art technology, which clearly identifies the specific finish or treatment to be used on the various components and assemblies, to provide protection against the various forms of materiel deterioration in the applicable operating and storage environments to which they may be exposed."

Custodians:
Army - ME
Navy - YD

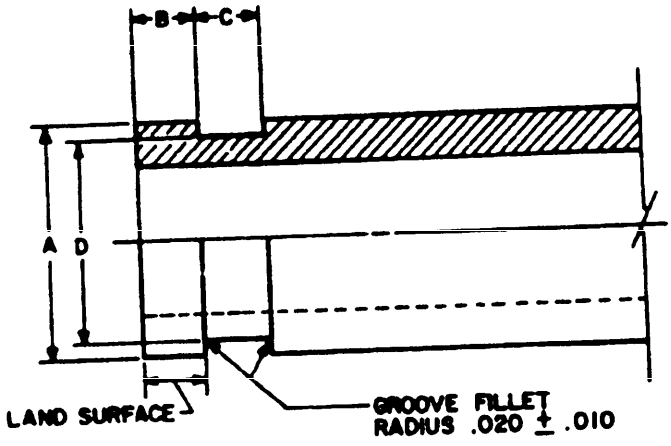
Preparing activity:
Army - ME

Review activity:
DLA - CS

User activity:
Navy - MC

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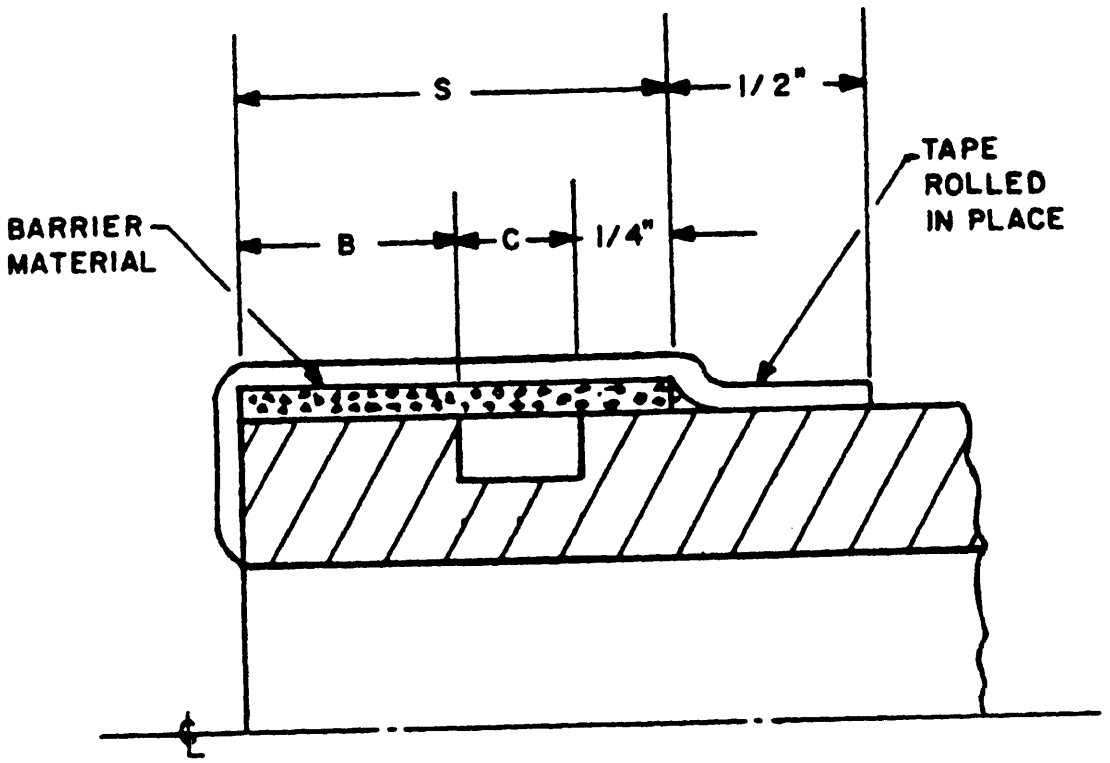
NOM PIPE SIZE	GROOVE DIMENSION (INCHES)				
	A		B	C	D
	ALUMINUM	STEEL, MALLEABLE IRON, OR DUCTILE IRON			
3/4	1.065 +0 −.016	1.050 +.031 −.016	0.625 ±.031	0.313 ± .031	0.938 +0.000 −0.015
1	1.330 +0 −.016	1.315 +.031 −.016	0.625 ±.031	0.313 ± 031	1.190 +0.000 −0.015
1-1/4	1.675 +0 −.016	1.660 +.031 −.016	0.625 ±.031	0.313 ± .031	1.535 +0.000 −0.015
1-1/2	1.915 +0 −.016	1.900 +.031 −.016	0.625 ±.031	0.313 ± .031	1.775 +0.000 −0.015
2	2.407 +0 −.016	2.375 +.031 −.016	0.625 ±.031	0.313 ± .031	2.250 +0.000 −0.015
2-1/2	2.908 +0 −.016	2.875 +.031 −.016	0.625 ±.031	0.313 ± .031	2.720 +0.000 −0.015
3	3.532 +0 −.016	3.500 +.031 −.016	0.625 ±.031	0.313 ± .031	3.344 +0.000 −0.018
3-1/2	4.040 +0 −.016	4.000 +.031 −.016	0.625 ±.031	0.313 ± .031	3.834 +0.000 −0.020
4	4.532 +0 −.016	4.500 +.031 −.016	0.625 ±.031	0.375 ± .031	4.334 +0.000 −0.020
5	5.620 +0 −.016	5.563 +.031 −.016	0.625 ±.031	0.375 ± .031	5.395 +0.000 −0.022
6	6.657 +0 −.016	6.625 +.031 −.016	0.625 ±.031	0.375 ± .031	6.455 +0.000 −0.022
8	8.657 +0 −.016	8.625 +.031 −.016	0.750 ±.031	0.437 ± .031	8.441 +0.000 −0.025
10	10.850 +0 −.032	10.750 +.047 −.031	0.750 ±.031	0.500 ± .031	10.562 +0.000 −0.027
12	12.797 +0 −.032	12.750 +.047 −.031	0.750 ±.031	0.500 ± .031	12.531 +0.000 −0.030
14	14.870 +0 −.032	14.000 +.047 −.031	0.938 ±.031	0.500 ± .031	13.781 +0.000 −0.030

NOTE:
ALL DIMENSIONS ARE IN INCHES

FIGURE 1. Dimensions for grooved ends for fittings with grooved backup shoulder.

X-2256A

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S ——— LAND SEALING SURFACE

B ——— LAND

C ——— GROOVE

FIGURE 2. Land sealing surface for grooved end pipe, tube end fittings.

X-449A

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-P-10388G Pipe Fittings: One or More Ends Grooved

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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